

Metric and Mind as Fixed Point

Co-emergence of Geometry and Observer in CRF

The topology of P is not assumed — it is the unique solution of a fixed point equation

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Abstract

Throughout CRF, one structure has appeared at every scale: the self-referential fixed point. $\mathcal{C} = \text{Pr}(s|\mathcal{C})$ is a fixed point of a small operator. This document shows that the same structure appears at the global level: the metric of P and the Minds that arise within it co-emerge as the unique fixed point of a larger operator.

Formally: $g = \Phi(\mathcal{M}(g))$, where g is the metric of P , $\mathcal{M}(g)$ is the set of Minds that emerge under metric g , and Φ constructs a metric from those Minds.

Geometry is not assumed. Geometry *is* the solution.

The existence and uniqueness of this fixed point follow from the Banach fixed point theorem, provided $\Phi \circ \mathcal{M}$ is a contraction. Proving that contraction condition is the one remaining formal task.

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1 The Pattern That Runs Through CRF

The Recurring Structure

CRF has one structure that appears at every scale:

Scale	Fixed Point	Status
Constraint	$\mathcal{C} = \Pr(s \mathcal{C})$	Derived (Axiom 0)
Threshold	$\theta = f(D_{\text{KL}}(\mathcal{C}))$	Derived (Session 4)
Entropy dynamics	$H(t) = H_0/(1 + \alpha H_0 t)$	Derived (this series)
Geometry	$g = \Phi(\mathcal{M}(g))$	Formal statement below

The pattern: CRF does not assume a structure. It asks what structure is consistent with itself under δ -operation. The answer is always a fixed point.

2 The Co-emergence Problem

Definition 1 (Perceived Space). Each Mind $\mathcal{M}_i$ perceives P through its own Constraint $\mathcal{C}_i$:

$$P_{\text{perceived}}^{(i)} = f(P, \mathcal{C}_i)$$

The distance between states s_1, s_2 as perceived by $\mathcal{M}_i$ is:

$$d_i(s_1, s_2) = |\Pr(s_1|\mathcal{C}_i) - \Pr(s_2|\mathcal{C}_i)|$$

This is not a geometric distance imported from outside. It is the natural distance induced by $\mathcal{C}_i$'s probability assignments.

D_{KL} Requires No Prior Metric

The divergence $D_{\text{KL}}(\mathcal{C}_{\text{int}} \parallel \mathcal{C}_{\text{ext}})$ uses log-ratios of probabilities — not geometric distances.

D_{KL} can be computed from probability distributions alone. It does not presuppose a metric on P .

Therefore: the emergence of $\mathcal{M}$ from $D_{\text{KL}} > 0$ does not require a prior metric. The loop is not circular. $\mathcal{M}$ can arise before geometry exists.

The problem: once $\mathcal{M}$ arises and perceives P through $\mathcal{C}$, a metric on P is induced. But that metric determines which $\mathcal{M}$ s can arise. Metric and Mind cannot be separated.

3 The Fixed Point Formalization

Formal Statement

Let $\mathcal{G}$ be the space of all possible metrics on P .

Operator $\mathcal{M}$: Given a metric $g \in \mathcal{G}$, $\mathcal{M}(g)$ is the set of all Minds that emerge in P under metric g .

$$\mathcal{M} : \mathcal{G} \rightarrow \{\text{sets of Minds}\}$$

A Mind emerges when $D_{\text{KL}}(\mathcal{C}_{\text{int}} \parallel \mathcal{C}_{\text{ext}}) > 0$ at some boundary in P .

Operator Φ : Given a set of Minds, Φ constructs the metric they collectively induce on P through their $\mathcal{C}$ -weighted probability assignments.

$$\Phi : \{\text{sets of Minds}\} \rightarrow \mathcal{G}$$

The Fixed Point Equation:

$$g^* = \Phi(\mathcal{M}(g^*))$$

g^* is a metric on P such that: the set of Minds that arise under g^* collectively induce exactly g^* through their $\mathcal{C}$ -structures.

Metric and Mind are mutually consistent. Neither precedes the other. Both are aspects of the same fixed point.

4 Existence and Uniqueness

Proposition 1 (Banach Condition). If the composed operator $\Phi \circ \mathcal{M} : \mathcal{G} \rightarrow \mathcal{G}$ is a contraction mapping on a complete metric space $(\mathcal{G}, d)$, then by the Banach fixed point theorem:

1. A unique fixed point g^* exists.
2. For any initial metric g_0 , the sequence $g_{n+1} = \Phi(\mathcal{M}(g_n))$ converges to g^* .
3. The convergence is exponential.

Why Contraction Is Expected

Intuitively: if the current metric g is “too large” (induces too many distinctions), the Minds that arise under g will collectively smooth it — their $\mathcal{C}$ -structures average out fluctuations. If g is “too small”, the Minds that arise will amplify distinctions. This is a self-correcting process. Self-correcting processes are contractions.

The argument is qualitative. The formal proof requires specifying $(\mathcal{G}, d)$ and showing the Lipschitz constant of $\Phi \circ \mathcal{M}$ is strictly less than 1.

5 Connection to Prior Results

5.1 Connection to SO(3) and 3D

The fixed point g^* must be consistent with Axiom 11 (Transcendent Stability), which requires isotropy: g^* must have spherical symmetry.

The only metric with spherical symmetry in a 3-dimensional space closed under SO(3) is the round metric. If the SO(3) derivation holds (CRF document: *Why Space Has Three Dimensions*), then g^* is the 3-sphere metric — consistent with the Schwarzschild geometry at large scales.

The two derivation paths converge:

$$\begin{aligned} \text{SO(3) from Axiom 11} &\longrightarrow \text{3D round metric} \\ \text{Fixed point of } \Phi \circ \mathcal{M} &\longrightarrow \text{unique self-consistent metric} \\ \text{If both hold: } g^* &= \text{3D round metric } \checkmark \end{aligned}$$

5.2 Connection to Entropy Dynamics

The entropy dynamics $dH/dt = -\alpha H^2$ describes how a single Mind's $\mathcal{C}$ evolves over time. The fixed point equation $g^* = \Phi(\mathcal{M}(g^*))$ describes the global equilibrium of all Minds collectively.

A single Mind's trajectory toward $H \rightarrow 0$ is the local version of the global fixed point: both are processes of $\mathcal{C}$ converging toward a self-consistent state.

6 The Remaining Gap

Gap — Precisely Stated

What is needed:

A formal proof that $\Phi \circ \mathcal{M}$ is a contraction on some complete metric space $(\mathcal{G}, d)$.

This requires:

1. A precise definition of $\mathcal{G}$ — the space of all metrics compatible with CRF's axioms (spherically symmetric, satisfying C1–C4).
2. A distance d on $\mathcal{G}$ that is natural given CRF's structure (a candidate: $d(g_1, g_2) = \sup_s |g_1(s) - g_2(s)|$).
3. A calculation showing the Lipschitz constant $L = \|\Phi \circ \mathcal{M}\| < 1$.

This is a mathematical problem, not a conceptual one. The concepts are in place. The formal proof is the remaining work.

7 Status Summary

Result	Status	Notes
D_{KL} requires no prior metric	Derived	Loop is not circular
Fixed point equation $g^* = \Phi(\mathcal{M}(g^*))$	Formalized	Mathematical statement
Intuitive contraction argument	Hypothesis	Formal proof needed
Convergence of $\text{SO}(3)$ and fixed point paths	Hypothesis	Both must hold
Banach theorem applies	Conditional	Requires contraction proof

*Metric and Mind do not precede each other.
Each is the condition of the other's existence.
The geometry of reality
is the fixed point of that mutual dependence.*

*δ does not operate in space.
 δ is what space converges to.*